PROBLEM 01

Rocket Helical Trajectory

Original statement · Problems.pdf, p. 6

MIT OCW Benchmark

An out-of-control rocket traces an upward helical trajectory described in Cartesian coordinates for $t \geq 0$ by:

$$x = a\cos(\omega t) \tag{1}$$

$$y = a\sin(\omega t) \tag{2}$$

$$z = v_z t \tag{3}$$

where a, v_z, and ω are constants.

(a) Express the position, velocity, and acceleration vectors in the Cartesian x, y, z system using unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.

(b) Express the position, velocity, and acceleration vectors in the cylindrical coordinate system using unit vectors $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$.

(c) Express the velocity and acceleration vectors using s, ρ and the vector basis $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$.

(d) Compute the modulus of the velocity vector.

(e) Express the evolution of the arc-length coordinate as a function of time.

(f) Write down the trajectory as a function of the arc-length coordinate.

(g) Where is the center of curvature for this curve when the particle is located at $(x, y, z) = (a, 0, 0)$?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** An out-of-control sounding rocket ascends following a regular circular helix around the vertical z-axis.
- **Hypotheses:** The cylinder radius $a > 0$, vertical ascent velocity $v_z > 0$, and angular spin rate $\omega > 0$ are strictly positive constants. The rocket is modeled as an ideal point particle P.
- **Configuration Degrees of Freedom (CDOFs):** The trajectory is purely kinematically prescribed by time t; thus, the particle has $\mathrm{CDOF} = 1$ parameterized by time t or arc length s.
- **Parameters & SI Units:** Radius a $[\mathrm{m}]$, angular frequency ω $[\mathrm{rad/s}]$, vertical velocity v_z $[\mathrm{m/s}]$, time t $[\mathrm{s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- **Inertial Reference Frame:** $\mathcal{S}_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, fixed to the launch site with vertical axis $\mathbf{k}_0$ aligned with z.
- **Cylindrical Moving Basis:** $\mathcal{B}_{\mathrm{cyl}} = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z\}$.

The transformation matrix from the inertial Cartesian basis $\mathcal{B}_0$ to the cylindrical basis $\mathcal{B}_{\mathrm{cyl}}$ is the planar rotation matrix about $\mathbf{k}_0$:

$$[{}_0R_{\text{cyl}}] = \begin{pmatrix} \cos(\omega t) & -\sin(\omega t) & 0 \\ \sin(\omega t) & \cos(\omega t) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The basis vector relationships are given by matrix-vector multiplication:

$$\begin{pmatrix} \mathbf{e}_R \\ \mathbf{e}_\theta \\ \mathbf{e}_z \end{pmatrix} = [{}_0R_{\text{cyl}}]^T \begin{pmatrix} \mathbf{i}_0 \\ \mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix} = \begin{pmatrix} \cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0 \\ -\sin(\omega t)\mathbf{i}_0 + \cos(\omega t)\mathbf{j}_0 \\ \mathbf{k}_0 \end{pmatrix}$$

- **Time Derivatives of Unit Vectors (Poisson's Theorem, Notes Sec. 3.1, Eqs. 3.3–3.5):**

Since $\mathcal{B}_{\text{cyl}}$ rotates at angular velocity $\boldsymbol{\omega} = \omega \mathbf{k}_0 = \omega \mathbf{e}_z$:

$$\frac{d\mathbf{e}_R}{dt} = \boldsymbol{\omega} \times \mathbf{e}_R = (\omega \mathbf{e}_z) \times \mathbf{e}_R = \omega \mathbf{e}_\theta$$

$$\frac{d\mathbf{e}_\theta}{dt} = \boldsymbol{\omega} \times \mathbf{e}_\theta = (\omega \mathbf{e}_z) \times \mathbf{e}_\theta = -\omega \mathbf{e}_R$$

$$\frac{d\mathbf{e}_z}{dt} = \mathbf{0}$$

- **Orthonormality of $[{}_0R_{\text{cyl}}]$:** $\det[{}_0R_{\text{cyl}}] = \cos^2(\omega t) + \sin^2(\omega t) = 1$ and $[{}_0R_{\text{cyl}}][{}_0R_{\text{cyl}}]^T = I$ (checked symbolically with sympy).
- **Frenet basis from the cylindrical basis.** The components of $\mathbf{e}_t$, $\mathbf{e}_n$, $\mathbf{e}_b$ in $\mathcal{B}_{\text{cyl}}$ (obtained in Phase 3, part (c)) are the columns of the second change-of-basis matrix, with $V = \sqrt{a^2\omega^2 + v_z^2}$:

$$[{}_{\text{cyl}}R_{\text{tnb}}] = \begin{pmatrix} 0 & -1 & 0 \\ a\omega/V & 0 & -v_z/V \\ v_z/V & 0 & a\omega/V \end{pmatrix}, \qquad \det = \frac{a^2\omega^2 + v_z^2}{V^2} = 1, \qquad RR^T = I$$

- The three columns have unit norm and are mutually orthogonal, so $RR^T = I$; the composition $[{}_0R_{\text{tnb}}] = [{}_0R_{\text{cyl}}][{}_{\text{cyl}}R_{\text{tnb}}]$ is again a proper rotation (determinant $1 \cdot 1 = 1$). Both determinants and the products RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Cartesian Position, Velocity, and Acceleration

Pedagogical Justification: The position vector $\mathbf{r}(t)$ is directly defined by its Cartesian coordinates $(x(t), y(t), z(t))$ relative to the fixed origin O. Velocity and acceleration are obtained by direct term-by-term differentiation in the fixed inertial basis $\mathcal{B}_0$, where $\frac{d\mathbf{i}_0}{dt} = \frac{d\mathbf{j}_0}{dt} = \frac{d\mathbf{k}_0}{dt} = \mathbf{0}$.

- **Position Vector:**

$$\mathbf{r}(t) = a\cos(\omega t)\mathbf{i}_0 + a\sin(\omega t)\mathbf{j}_0 + v_z t\mathbf{k}_0$$

- **Velocity Vector (Explicit Chain Rule):**

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \frac{d}{dt}[a\cos(\omega t)]\mathbf{i}_0 + \frac{d}{dt}[a\sin(\omega t)]\mathbf{j}_0 + \frac{d}{dt}[v_z t]\mathbf{k}_0$$

Applying the chain rule $\frac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$ and $\frac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$:

$$\mathbf{v}(t) = -a\omega\sin(\omega t)\mathbf{i}_0 + a\omega\cos(\omega t)\mathbf{j}_0 + v_z\mathbf{k}_0$$

- **Acceleration Vector (Explicit Chain Rule):**

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d}{dt}[-a\omega\sin(\omega t)]\mathbf{i}_0 + \frac{d}{dt}[a\omega\cos(\omega t)]\mathbf{j}_0 + \frac{d}{dt}[v_z]\mathbf{k}_0$$

$$\mathbf{a}(t) = -a\omega^2\cos(\omega t)\mathbf{i}_0 - a\omega^2\sin(\omega t)\mathbf{j}_0 + 0\mathbf{k}_0 = -a\omega^2(\cos(\omega t)\mathbf{i}_0 + \sin(\omega t)\mathbf{j}_0)$$

(b) Cylindrical Position, Velocity, and Acceleration

> **Pedagogical Justification:** Cylindrical coordinates (R, θ, z) exploit the axial symmetry of the cylinder upon which the helix lies. Here, the radial distance $R(t) = \sqrt{x^2+y^2} = a$ is constant, and the azimuthal angle is $\theta(t) = \omega t$.

- **Position in Cylindrical Coordinates:**

$$\mathbf{r}(t) = a\mathbf{e}_R + v_z t\mathbf{e}_z$$

- **Velocity Vector (Explicit Product Rule with Poisson Derivative):**

$$\mathbf{v}(t) = \frac{d}{dt}(a\mathbf{e}_R + v_z t\mathbf{e}_z) = a\frac{d\mathbf{e}_R}{dt} + \frac{d(v_z t)}{dt}\mathbf{e}_z + v_z t\frac{d\mathbf{e}_z}{dt}$$

Substituting $\frac{d\mathbf{e}_R}{dt} = \omega\mathbf{e}_\theta$ and $\frac{d\mathbf{e}_z}{dt} = \mathbf{0}$:

$$\mathbf{v}(t) = a\omega\mathbf{e}_\theta + v_z\mathbf{e}_z$$

- **Acceleration Vector (Explicit Differentiation):**

$$\mathbf{a}(t) = \frac{d}{dt}(a\omega\mathbf{e}_\theta + v_z\mathbf{e}_z) = a\omega\frac{d\mathbf{e}_\theta}{dt} + 0 = a\omega(-\omega\mathbf{e}_R) = -a\omega^2\mathbf{e}_R$$

(c) Intrinsic Frenet-Serret Basis $\{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$

> **Pedagogical Justification:** The Frenet-Serret intrinsic basis (Notes Sec. 2.4) decomposes motion naturally along the curve itself. The tangent vector $\mathbf{e}_t$ points in the direction of instantaneous velocity, the principal normal $\mathbf{e}_n$ points toward the center of curvature, and the binormal $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$ completes the right-handed orthonormal triad.

- **Unit Tangent Vector $\mathbf{e}_t$:**

$$\mathbf{e}_t = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}}$$

- **Unit Principal Normal Vector $\mathbf{e}_n$:**

By definition of the intrinsic acceleration (Notes Eq. 2.17):

$$\mathbf{a} = \ddot{s}\mathbf{e}_t + \frac{\dot{s}^2}{\rho}\mathbf{e}_n$$

Since the speed $v = \dot{s} = \sqrt{a^2\omega^2 + v_z^2}$ is strictly constant in time, the tangential acceleration vanishes:

$$\ddot{s} = \frac{dv}{dt} = 0$$

Therefore, the entire acceleration is purely normal:

$$\mathbf{a} = a_n \mathbf{e}_n = -a\omega^2 \mathbf{e}_R$$

Taking the norm of both sides yields the magnitude of normal acceleration:

$$a_n = \|\mathbf{a}\| = a\omega^2$$

Dividing $\mathbf{a}$ by its norm gives the unit normal vector:

$$\mathbf{e}_n = \frac{\mathbf{a}}{a_n} = \frac{-a\omega^2 \mathbf{e}_R}{a\omega^2} = -\mathbf{e}_R$$

The principal normal points radially inward toward the central axis of the cylinder!

- **Radius of Curvature ρ:**

Using the relation $a_n = \frac{v^2}{\rho}$:

$$\rho = \frac{v^2}{a_n} = \frac{a^2\omega^2 + v_z^2}{a\omega^2} = a + \frac{v_z^2}{a\omega^2}$$

- **Unit Binormal Vector $\mathbf{e}_b$:**

Computing the vector cross product in the cylindrical basis:

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n = \left(\frac{a\omega \mathbf{e}_\theta + v_z \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}} \right) \times (-\mathbf{e}_R) = \frac{-a\omega(\mathbf{e}_\theta \times \mathbf{e}_R) - v_z(\mathbf{e}_z \times \mathbf{e}_R)}{\sqrt{a^2\omega^2 + v_z^2}}$$

Recalling cyclic cross products: $\mathbf{e}_\theta \times \mathbf{e}_R = -\mathbf{e}_z$ and $\mathbf{e}_z \times \mathbf{e}_R = \mathbf{e}_\theta$:

$$\mathbf{e}_b = \frac{-a\omega(-\mathbf{e}_z) - v_z \mathbf{e}_\theta}{\sqrt{a^2\omega^2 + v_z^2}} = \frac{-v_z \mathbf{e}_\theta + a\omega \mathbf{e}_z}{\sqrt{a^2\omega^2 + v_z^2}}$$

(d) Modulus of Velocity

> **Pedagogical Justification:** The speed $v(t) = \|\mathbf{v}(t)\|$ represents the scalar rate of distance traveled along the path.

- **Calculation:**

$$v = \|\mathbf{v}\| = \sqrt{(-a\omega \sin \omega t)^2 + (a\omega \cos \omega t)^2 + v_z^2} = \sqrt{a^2\omega^2(\sin^2 \omega t + \cos^2 \omega t) + v_z^2} = \sqrt{a^2\omega^2 + v_z^2}$$

The speed is constant for all $t \geq 0$.

(e) Evolution of Arc Length Coordinate $s(t)$

> **Pedagogical Justification:** The arc-length coordinate $s(t)$ measures the distance traversed along the trajectory from the initial position at $t = 0$. By fundamental calculus:

$$s(t) = \int_0^t v(t')\, dt'$$

- **Definite Integration:**

$$s(t) = \int_0^t \sqrt{a^2\omega^2 + v_z^2}\, dt' = \sqrt{a^2\omega^2 + v_z^2}\, [t']_0^t = t\sqrt{a^2\omega^2 + v_z^2}$$

(f) Trajectory as a Function of Arc Length s

> **Pedagogical Justification:** To express the curve intrinsically independent of time, we invert $s(t)$ to obtain time as a function of arc length: $t(s) = \frac{s}{\sqrt{a^2\omega^2 + v_z^2}}$.

- **Substitution into Cartesian Coordinates:**

$$x(s) = a \cos\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right)$$

$$y(s) = a \sin\left(\frac{\omega s}{\sqrt{a^2\omega^2 + v_z^2}}\right)$$

$$z(s) = \frac{v_z s}{\sqrt{a^2\omega^2 + v_z^2}}$$

(g) Center of Curvature C at $(a, 0, 0)$

> **Pedagogical Justification:** The center of curvature C of the osculating circle is displaced from particle P along the principal normal $\mathbf{e}_n$ by the radius of curvature ρ (Notes Sec. 2.4, Fig. 2.2):

$$\mathbf{r}_C = \mathbf{r}_P + \rho \mathbf{e}_n$$

- **Evaluation at $(a, 0, 0)$:**

At position $(a, 0, 0)$, $x(t) = a\cos(\omega t) = a \implies \omega t = 0 \implies t = 0$.

At $t = 0$:

- Position: $\mathbf{r}_P(0) = a\mathbf{i}_0$.
- Unit normal: $\mathbf{e}_n(0) = -\mathbf{e}_R(0) = -(\cos(0)\mathbf{i}_0 + \sin(0)\mathbf{j}_0) = -\mathbf{i}_0$.
- Radius of curvature: $\rho = a + \frac{v_z^2}{a\omega^2}$.
- **Algebraic Combination:**

$$\mathbf{r}_C = a\mathbf{i}_0 + \left(a + \frac{v_z^2}{a\omega^2}\right)(-\mathbf{i}_0) = a\mathbf{i}_0 - a\mathbf{i}_0 - \frac{v_z^2}{a\omega^2}\mathbf{i}_0 = -\frac{v_z^2}{a\omega^2}\mathbf{i}_0$$

In Cartesian coordinates:

$$C = \left(-\frac{v_z^2}{a\omega^2},\, 0,\, 0 \right)$$

PHASE 4

Physical Interpretation & Dimensional Verification

- **Dimensional Check:**
- Velocity: $[a\omega] = [\mathrm{m}][\mathrm{s}^{-1}] = \mathrm{m/s}$, $[v_z] = \mathrm{m/s}$. $\implies$ $\sqrt{a^2\omega^2 + v_z^2}$ has units $\mathrm{m/s}$ (Consistent).
- Curvature center offset: $\left[\frac{v_z^2}{a\omega^2}\right] = \frac{[\mathrm{m/s}]^2}{[\mathrm{m}][\mathrm{s}^{-1}]^2} = \frac{\mathrm{m}^2/\mathrm{s}^2}{\mathrm{m/s}^2} = \mathrm{m}$ (Consistent).
- **Limiting Asymptotic Cases:**
- If $v_z \to 0$ (planar circular motion): $\rho = a$, and $C = (0, 0, 0)$, the origin of the circle as expected physically.
- If $\omega \to 0$ (pure vertical line ascent): $\rho \to \infty$, curvature vanishes, and the trajectory becomes a straight vertical line.

Source: Problems.pdf (Taken from MIT OCW)